

Desk Caddy Design & 3D Printing Project

Overview

In this project, you will design a **functional desk caddy** using Autodesk Inventor. Your desk caddy should be designed to organize common desk supplies such as pens, pencils, scissors, sticky notes, paper clips, and other small items.

You will begin by researching existing desk organizers and identifying a design that you like. You will then use that research as inspiration to develop your own original design in Inventor.

The final goal is to create a desk caddy that can be **3-D printed and used in the real world**. Due October 23rd.

Design Challenge

Design and model a 3-D printable desk caddy that organizes at least three different types of desk supplies.

Your design should balance:

- Functionality
- Appearance
- Ergonomics
- Efficient use of material
- Ease of 3-D printing
- Creativity

Your finished caddy should be something you would actually want to use on a desk.

Part 1 — Research

Before beginning your design, research existing desk organizers.

Find **at least two examples** of desk caddies or organizers online.

For each example, record:

1. A picture or link to the product
2. Approximate overall dimensions
3. What items it is designed to hold
4. Features you like
5. Features you would change
6. What ideas you could incorporate into your own design

Research Question

After looking at existing products, answer:

What features make a desk organizer useful and easy to use?

Your answer should be at least one paragraph.

Part 2 — Design Requirements

Your desk caddy must meet the following minimum requirements.

Required Features

- Hold **pens and/or pencils**
- Hold **scissors or another tall desk tool**
- Include a compartment or way to store **sticky notes**
- Include at least **one additional storage area** for items such as:
 - Paper clips
 - Erasers
 - USB drives
 - Binder clips
 - Other small objects

Size

The finished caddy should be approximately:

- **Maximum width:** 220 mm (approximately 8.5 inches)
- **Maximum depth:** 120 mm (approximately 4.75 inches)
- **Maximum height:** 120 mm (approximately 4.75 inches)

3-D Printing Requirements

The design should be suitable for FDM 3-D printing.

Consider:

- Wall thickness
- Overhangs
- Supports
- Print orientation
- Material usage
- Strength
- Stability
- Build-plate size

Your design should be able to sit securely on a desk without tipping over easily.

Part 3 — Sketching & Planning

Before creating your final model, produce at least **two concept sketches** (these can be by hand or any suitable digital drawing tool)

Each sketch should show:

- Overall shape
- Major compartments
- Approximate dimensions
- What each compartment will hold

Label important dimensions and features.

After reviewing your concepts, select one design to develop in Inventor.

Be prepared to explain why you selected it.

Part 4 — Autodesk Inventor Model

Create your desk caddy as a 3-D model in Autodesk Inventor.

Your model should demonstrate appropriate use of Inventor modeling techniques.

Suggested techniques

You may use:

- 2-D sketches
- Extrude
- Cut
- Fillet
- Chamfer
- Shell
- Patterns
- Construction geometry
- Parameters
- Constraints

Your model should be **fully constrained where appropriate** and use logical dimensions (for example 124.58 mm is not logical)

Recommended starting dimensions

As a starting point, you may consider a caddy approximately:

200 mm × 100 mm × 100 mm

These dimensions are not required. Your design should ultimately be based on your own decisions and the needs of the objects it will hold.

Part 5 — Design for Manufacturing

(this will be adjusted if we don't get access to the printer)

Before submitting your model, evaluate whether it can actually be manufactured using a 3-D printer.

Consider the following:

Wall Thickness

Walls should be thick enough to survive normal use but not unnecessarily thick.

Stability

The caddy should have a sufficiently large base and should not easily tip when objects are placed in it.

Printability

Avoid unnecessary:

- Large unsupported overhangs
- Extremely thin walls
- Tiny details
- Difficult-to-print internal geometry

Material Efficiency

Think about how much plastic your design uses.

A successful design is not necessarily the one that uses the most material.

Part 6 — Prototype & Test

If a 3-D printer is available, print your design. I am hoping we can make this work and 3-D print your design

Test the finished caddy using actual desk supplies.

Evaluate:

- Does it hold the intended objects?
- Are the compartments large enough?
- Is it stable?
- Are the objects easy to remove?

- Is it comfortable to use?
- Is the design strong enough?

- Does it look like the intended design?
Record at least **three observations** about your prototype.
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Part 7 — Improve Your Design

Based on your testing, identify at least **two improvements** you would make to the design. If time and equipment allow, revise your Inventor model and create an improved version.

Examples:

- Increase the size of a compartment
 - Make a compartment deeper
 - Add rounded edges
 - Increase stability
 - Reduce material
 - Improve the appearance
 - Make a feature easier to use
 - Reduce the need for printing supports
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Required Deliverables

Submit the following:

1. Research

- Two reference products
- Images/links
- Approximate dimensions
- Written analysis

2. Concept Sketches

- Minimum of two designs
- Dimensions and labels

- Selected final concept

3. Inventor Model

Submit the required Inventor file(s).

Your final model should be complete, organized, and correctly dimensioned.

4. Drawing

Create a dimensioned technical drawing showing:

- Overall dimensions
- Important compartment dimensions
- At least three additional relevant dimensions
- Appropriate views

5. Design Reflection

Answer the following:

1. What problem does your desk caddy solve?
2. Who is the intended user?
3. What existing products influenced your design?
4. What was the most difficult part of the design?
5. What Inventor tools did you use?
6. How did you design the caddy for 3-D printing?
7. What did you learn from testing your prototype?
8. What would you change in a second version?

Grading Rubric — 100 Points

Category	Excellent	Proficient	Developing	Beginning	Points
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Research & Inspiration	Thorough research of 2+ products; dimensions and design features analyzed; clear connection to final design	Research completed with useful observations	Limited research or incomplete analysis	Little/no meaningful research	10
Concept Development	2+ thoughtful concepts with clear dimensions and labels; design decisions are well justified	Required concepts completed with reasonable planning	Concepts are incomplete or lack detail	Little evidence of planning	10
Functionality	Effectively organizes 4+ categories of items; compartments are appropriately sized and easy to use	Successfully meets all required storage functions	Meets most requirements but has usability issues	Does not meet major requirements	20
Inventor Modeling	Accurate, well-constructed model; excellent use of sketches, constraints, features, and dimensions	Complete model with appropriate Inventor techniques	Model is functional but contains modeling/design issues	Model is incomplete or poorly constructed	20
Design for 3-D Printing	Excellent consideration of wall thickness, stability, orientation, supports, strength, and material use	Design is clearly printable with minor issues	Some significant printing concerns	Design is not realistically printable	15

Creativity & Aesthetics	Original, thoughtful design with strong visual appeal and purposeful details	Attractive design with some original features	Basic design with limited creativity	Minimal attention to appearance or originality	10
Prototype & Testing	Thorough testing with specific observations and meaningful improvements	Prototype tested and improvements identified	Limited testing or vague observations	Little/no testing	5
Technical Drawing	Accurate, complete, properly dimensioned drawing using appropriate views	Complete drawing with minor errors	Missing dimensions/views or several errors	Incomplete drawing	5
Reflection & Communication	Clear, thoughtful reflection demonstrating strong understanding of the design process	Adequate reflection with clear responses	Incomplete or superficial reflection	Minimal reflection	5
Total					100

Student Success Checklist

Before submitting, make sure you can answer **yes** to each question:

- I researched at least two existing desk organizers.
- I documented the approximate dimensions of my reference designs.
- I created at least two concept sketches.

- My caddy holds pens/pencils.
- My caddy holds scissors or another tall tool.
- My caddy has a sticky-note compartment.
- My caddy has an additional storage compartment.
- My design is within the assigned size limits - -you can go over a little in either direction.
- My Inventor model is complete.
- My design is suitable for 3-D printing.
- I created a dimensioned technical drawing.
- I tested my design or considered how it would be tested.
- I identified improvements for a future version.
- I completed my design reflection.

Extension Challenge

For an additional challenge, modify your desk caddy to include one or more advanced features:

- Modular/removable compartments
- Interlocking components
- Stackable caddies
- A phone stand
- Cable management
- A ruler slot
- Magnetic inserts
- A hidden compartment
- Custom name/logo
- Parametric dimensions that allow the caddy to be resized
- A design that can be printed without supports

Your extension should serve a **functional purpose** rather than simply adding decoration.

Final Design Question

If you were going to manufacture 100 of your desk caddies, what would you change about your design to make it cheaper, faster, stronger, or easier to manufacture?